

Data Storytelling



Module 3: Dashboards & Stories in Tableau

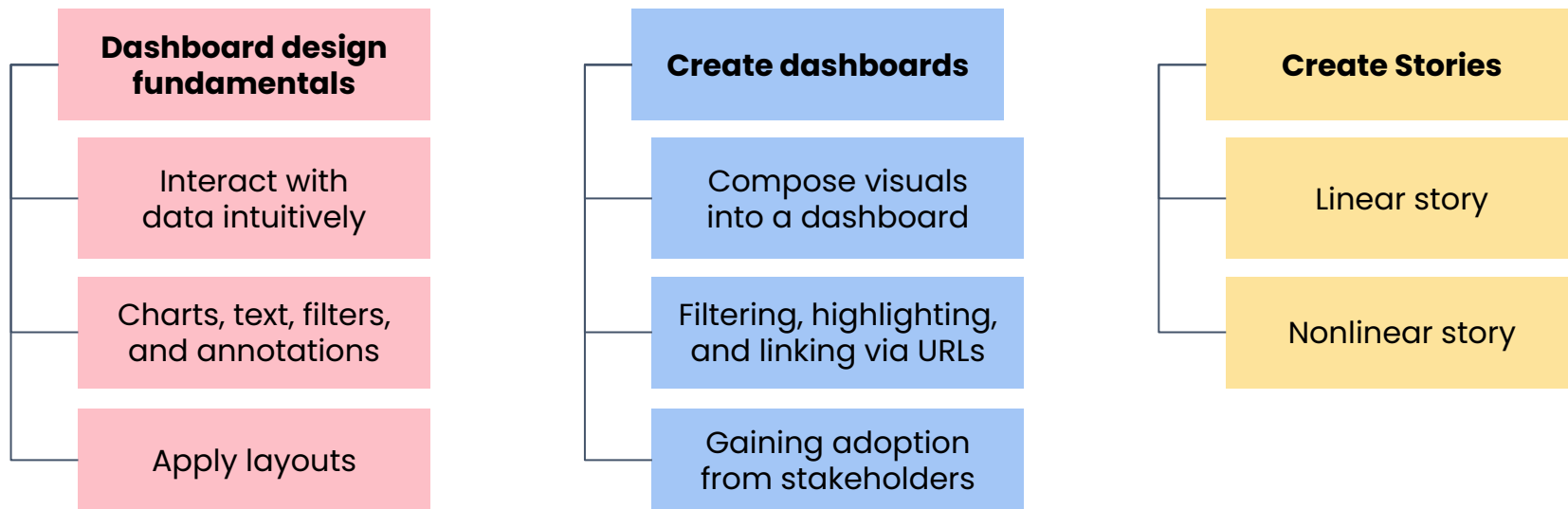




Dashboards & Stories in Tableau

Module 3 introduction

Module 3 outline



Dashboards & Stories in Tableau

Introduction to
dashboards

What is a dashboard?

- An interactive display of many related charts
- Gathers all relevant information in one place
- Central hub where audience can explore insights on their own
 - ✓ Allows stakeholders more autonomy
 - ✓ Explore data at their own pace
 - ✓ View insights that matter most to them
 - ✓ Draw their own conclusions
- Typically hooked to dynamic data
 - **Example dashboard:** Outdoor equipment sales might be refreshed every 10 mins with up-to-date sales information
- Tableau allows you to:
 - Connect to dynamic sources
 - Display most current information

Dashboard views

- Organize information in different ways, while keeping insights grounded in the same data.
- You can design multiple entry points tailored to different audiences.
- **Example:** Separate views for executives and product managers
 - Tailoring structure for each group
 - Keeping overall insights the same
 - Save time, since you're reusing same charts and data connections

Built-in features

 **Cross-filtering:** Connects multiple charts so selecting data in one filters data in another.

- Map displaying countries in Asia
- Donut chart of sales by category
- User clicks on country → Donut chart is filtered to show sales in country

 **Actions:** Trigger navigation between pages or take user to external links:

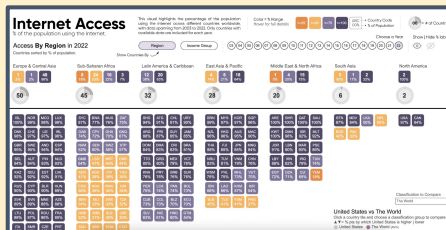
- Clicking category in donut chart might highlight same category in line graph
- Clicking link in tooltip might take user to listing for a given product.

 **Dynamic text and titles:** Update automatically based on user selections.

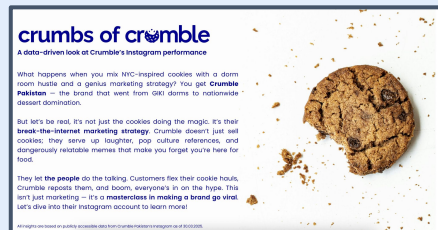
- User selects "India" from the map,
- Title automatically changes from "Asia Sales Overview" to "India Sales Overview,"
- Summary text could update to "India accounts for 23% of regional revenue"

Dashboards & Stories in Tableau

Components of a
dashboard



- Many related charts that don't tell a single story
- Allow user to locate insights
- More information dense and answers a greater variety of questions
- Autonomy & intention



- Intended to guide user through specific narrative.
- Answers one core question:
 - "What makes Crumble Cookies' Instagram successful?"
- Easier to use for stakeholders, because there's one path

Weakly curated

Strongly curated

Dashboards & Stories in Tableau

Selecting insights for a
dashboard

Know your audience

Audience	Needs
Technical users	• Complex visualizations and filtering options • Drill into details, manipulate variables, explore relationships
Executives	• High-level metrics with more curated data story • Clear visualizations that highlight key performance indicators
Your team	• Detailed operational metrics that help with day-to-day decision making
You	• Prioritize functionality over aesthetics • More complex and less polished

Steps to evaluate target audience

1. Assess stakeholders' data literacy
 - Are they familiar with statistical testing?
 - Are they comfortable interpreting a chart with multiple dimensions?
2. Identify the decisions they need to make
3. Identify their time constraints
 - Do they tend to have 30 seconds or 30 minutes?
 - Will you be able to walk them through dashboard their first time using it?

Effective dashboards

- ✓ Built with a clear purpose in mind
- ✓ Answer specific questions like:
 - "How did Crumble Cookie's Instagram achieve virality?"
 - "What outdoor products are bringing in the most revenue?"
- ✗ Without a clear objective:
 - Dashboard risks becoming cluttered and ineffective

Example: Improvements in patient outcomes

- Select more curated story
- Explain:
 - How much outcomes have improved
 - What factors contributed to outcome

Example: Evaluating potential content investments

- Create an exploration sandbox where stakeholders can investigate questions:
 - "Which genres consistently deliver on time?"
 - "How do production budgets correlate with viewer ratings?"

Focus on decisions

Most effective dashboards are **decision-support** tools:



Ask yourself

- How will this chart or data point drive action?
- Does this metric address the problem we're trying to solve?
- Is this information necessary, or just nice to have?



More elements → More you risk diluting that objective

- Each component should contribute to a cohesive whole
- Good practice to:
 - ✓ Write out specific question each visualization answers
 - ✓ Evaluate whether question aligns with dashboard's purpose

Tips to guide attention

- Provide clear guidance on what the audience should focus on
- Consider the hierarchy of information
- Primary metrics that directly inform key decisions should be the most prominent
- Supporting information can be:
 - De-emphasized
 - Available when actively searched for



Dashboards & Stories in Tableau

Dashboards: the good
& the better

Dashboard layout fundamentals

Chart placement

- ☐ Placing most important metrics at top or in center
- ☐ Use common templates to help organize information logically
- ☐ Group related visuals together

Chart size

- ☐ Size should depend on the type of data they present
- ☐ Avoid overcrowding with too many small charts

Consistent structure

- ☐ Make title descriptive and relevant
- ☐ Use explanatory text to help users understand what dashboard shows
- ☐ Arrange views of charts and visuals
 - Highly structured story → Arrange from high level to details
 - Sandbox-style → Cluster related visuals together
- ☐ Add filters and actions

Ineffective dashboards

- Don't answer core business question
- Fail to connect with target audience

🔍 Identified by:

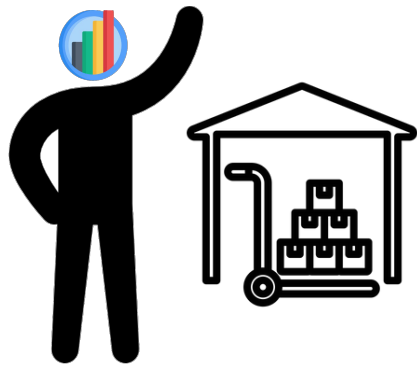
- ☐ **Cluttered layout:** Too many visuals and text without a clear hierarchy
- ☐ **Inconsistent design:** Clashing colors, inconsistent fonts and chart styles
- ☐ **Irrelevant metrics:** Data doesn't align with user's goals or charts don't contribute to core story
- ☐ **Poor visualization design:** Using pie charts with too many categories or unlabeled axes



Dashboards & Stories in Tableau

Creating a dashboard
in Tableau

Scenario



You

Data Analyst



Company: Northwind Traders



Task: Create a dashboard for the sales team to help them explore the countries and companies with the most orders



Audience: Sales team

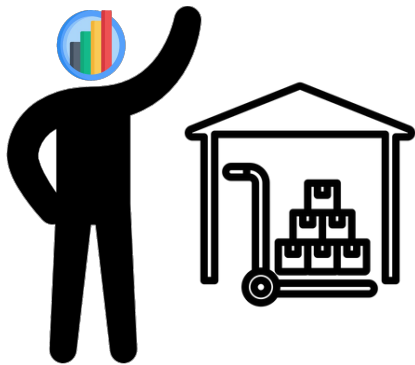
- Some are highly data literate
- Others are less comfortable with interpreting insights



Business questions:

- “Which countries generate the most orders?”
- “Which companies are buying the most products?”
- “Which countries or companies should we consider closing our contracts with this coming season?”

Scenario



You
Data Analyst

- Dashboard will require **moderate** curation
- Audience has many different but related questions
 - Need a moderate degree of autonomy to identify the relevant insights.
 - Organize charts and tables into a logical flow that helps walk users through insights



Plan: Create a dashboard with three visualizations

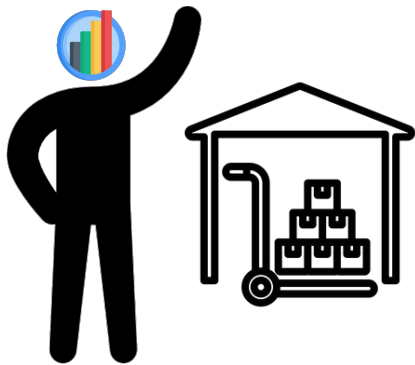
- ☐ Map of orders by country
- ☐ Chart of order features by country and company
- ☐ Time series chart



Dashboards & Stories in Tableau

Layout design

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create a dashboard for the sales team to help them explore countries and companies with most orders

Dashboards & Stories in Tableau

Actions

Actions in Tableau

Actions add interactivity to dashboards:



Filter: Update charts dynamically based on selections made in other charts

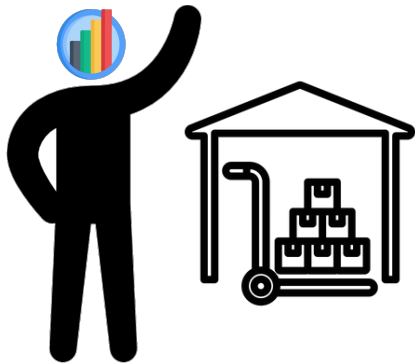


Highlight: Used to emphasize related data points across different charts



Go to URL : Enables linking to external websites

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create a dashboard for the sales team to help them explore countries and companies with most orders

Dashboards & Stories in Tableau

Publishing a dashboard

Different views

- Stakeholders will be viewing dashboard from a variety of devices, including:



Laptops



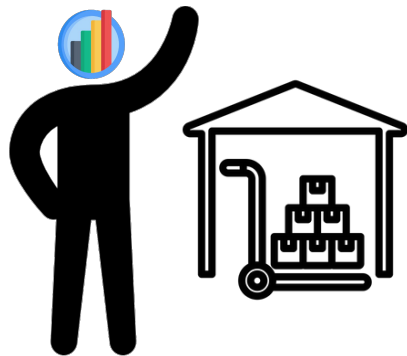
Widescreen monitors



Phones

- Make sure dashboard looks good on a **laptop** and a **widescreen monitor**
 - ❌ Won't account for every device
 - ✅ Make sure dashboard gains adoption

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create a dashboard for the sales team to help them explore countries and companies with most orders



Preview final dashboard on different devices and publish it



Dashboards & Stories in Tableau

Gaining adoption for
your dashboard

Dashboard adoption

- Dashboard is only as valuable as the number of people actively using it
- If dashboard isn't seeing consistent use:
 - ❌ Content doesn't address needs
 - ❌ Stakeholders don't understand how to navigate it
- Avoid assuming that users will instantly see the value of your dashboard
 - ✅ "Getting started" or walkthrough can help overcome learning curve
 - ✅ Identify points of confusion and address with clear instructions

Methods for gaining adoption



Documentation:

- Written guides or text pop-ups explaining components
- Give users a reference point for how to interact or interpret insights



Training:

- Hold a live session
- Show how to:
 - Drill into specific data points
 - Set meaningful filters
 - Exporting information



Video tutorial:

- Screen-record a quick tour
- Explain each major section and how it supports common use cases
- Helpful for new team members

Onboarding

- Never assume users will instantly see the value of your dashboard
- Short “getting started” section or interactive walkthrough can help new visitors:
 - ✓ Overcome initial learning curve
 - ✓ Feel comfortable navigating filters, customizing views, and extracting data
 - ✓ Be more likely to use the dashboard

Measuring success

- Tableau Public provides usage statistics that measure engagement
- Tableau Public metrics include:
 - Total views
 - Number of times visualizations were favorited
- Tableau Desktop includes:
 - Number of unique visitors
 - Time spent viewing dashboard
 - Interactions with filters & charts

Example: Certain visualizations receive high engagement while others are rarely clicked



Assess whether truly underutilized or if they're not necessary



Ask if they contribute to the core data story and help answer questions

■ If valuable, try:

- Moving to more prominent position
- Adjusting visual design
- Making clearer with annotations

■ Otherwise, consider removing it.



Dashboards & Stories in Tableau

Creating a linear story

Tableau Stories

- A Tableau Story
 -  Collection of pages (called story points)
 -  Build up using charts or dashboards
- Each Story point can hold:
 - Single chart
 - Dashboard
- You can:
 - ☐ Create **narrative-driven** Story that guides your stakeholders through each key point
 - ☐ Create **exploration sandbox** with different views for different stakeholders



Dashboards & Stories in Tableau

Creating a nonlinear story

Nonlinear Stories

- Collections of visualizations and dashboards that don't conform to a narrative
- They might serve as:
 - ▣ An exploration sandbox
 - Allows audience to collect insights from variety of charts
 - ▣ Centralized hub for stakeholders
 - View insights tailored for them
 - Peek at what teams are interested in
 - Provide value while saving time

Example

- **Company:** Northwind Traders
- Hub for sales and supply chain teams
- Takes advantage of the visualizations built throughout this course



Dashboards & Stories in Tableau

Introduction to dashboards

What is a dashboard?

- A centralized, interactive canvas where all your most useful charts and metrics co-exist
- Give your audience a snapshot of data
- A central hub where they can:
 - ✓ Quickly compare data points
 - ✓ Uncover interesting relationships
 - ✓ Draw their own conclusions
- Valuable in fast-paced environments where quick decision-making is essential

Dashboard vs static report

Static report

- Captures data at one moment in time
- Shows story the way you set it up
- Only tells the story on page or screen
- If you had to produce a customized static report for every department, you'd quickly burn out

Dashboard

- Introduces interactivity
- Explores aspects static report might overlook
- Shifts from a linear story to open exploration
- Multiple entry points for different audiences
- Empowers each department to ask and answer their own questions

Why interactivity matters



At the heart of a great dashboard

- Tooltips reveal hidden details or precise values
- Filters drill down into specific segments
- Each user uncovers insights that matter to them



Don't have the same level of control over:

- How people interpret each nuance
- Exact sequence of insights

Your role

- ☐ Provide framework
- ☐ Ensure data is accurate
- ☐ Give user freedom to navigate
- ☐ Share responsibility of guiding narrative
- ☐ Design environment that's intuitive

Built-in features



Data Tables

- Integrate raw numbers into appendix
- For those who need underlying data



Tooltips

- Provide precise values, definitions, or context when hovering over elements



Tabbed visualizations

- Offer selectable options where each tab can focus on different aspects